

COMPUTER FORENSICS REPORT

INTERNATIONAL CYBERSECURITY & DIGITAL FORENSICS ACADEMY (ICDFA)

Case Study 3 – Illegal File Transferring: Memory, USB and User-Activity Forensics

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1. Student Declaration

I, Gabriel Ighietsemhe, declare that this report for CIP-B105 is my original work. All findings and conclusions are based on my independent examination of the provided evidence. I have not copied conclusions from external sources or walkthroughs; historical values were used solely to validate my independently generated results.

I adhered strictly to all safety and authorization guidelines. All analysis was performed on read-only or verified working copies to preserve digital integrity. No recovered accounts were accessed, no malware was executed, and no evidence was uploaded to public services.

In accordance with the institution's Responsible AI-use policy, an AI tool was used to assist with report structuring and drafting. However, the forensic analysis, technical findings, and final conclusions remain entirely my own human-led effort.

2. Executive Summary

This forensic examination was conducted under the authorization of the International Cybersecurity & Digital Forensics Academy (ICDFA) for the CIP-B105 Case Study 3 assessment (CIP-B105-CS3-17288). The primary objective was to perform a forensically sound, evidence-driven investigation into a Windows 7 32-bit memory image (memdumpWin7.mem) to test the hypothesis that a suspect downloaded a specific file and illegally transferred it to removable USB storage.

Master evidence files were write-protected upon ingestion, and integrity baselines were established. The investigative framework utilized a customized Volatility 2 workflow (vol.py) targeting several key evidence domains:

- **Memory Preservation & Environment Baseline:** Profiled the volatile memory dump as Win7SP1x86_23418 and mapped kernel structures.
- **Suspect Accounts & Registry Hives:** Enumerated loaded Registry hives in memory (SYSTEM, SOFTWARE, SAM, NTUSER.DAT) to identify active SIDs and user profiles.
- **System & Hardware Profile:** Reconstructed the target PC's hostname and configuration metrics from memory-resident Registry keys.
- **Network Objects & Process Forensics:** Scanned network objects (netscan) to isolate active and historical connection endpoints mapped to running process identifiers.
- **Command & Browser Activity:** Targeted command-line execution buffers (consoles) and browser memory structures to locate evidence of file downloading, staging, and copying actions.
- **USB & Shellbag Artifact Attribution:** Constructed the complete hardware attribution chain across USBSTOR, MountedDevices, and MountPoints2 keys, correlating findings with user folder navigation history.

3. Authority, Scope, and Investigative Questions

3.1 Legal Authority and Mandate

This examination is conducted under the authorization of the International Cybersecurity & Digital Forensics Academy (ICDFA) for the CIP-B105 Case Study 3 assessment. The objective is to conduct a forensically sound, evidence-driven investigation into a Windows memory image to determine whether a suspect downloaded a file and transferred it to removable storage, evaluating evidence across volatile memory, user accounts, network objects, USB attachment records, and shell artifacts.

3.2 Scope of Examination

The examination is restricted to the non-destructive analysis of the supplied memory image within the protected ICDFA laboratory environment. In accordance with course safety guidelines, analysis was performed exclusively on verified working copies. No external entities were contacted, credentials were not reused, and no live malware or unverified binaries were executed.

The examination specifically encompassed the following evidence domains:

- Memory Image Preservation, Hashing Control, and Volatility 2 Environment Setup
- Suspect Account Enumeration and Memory-Resident Registry Hive Parsing
- Suspect PC Hardware Profile and Device Enumeration
- Network Object Mapping, Process Forensics, and Suspicious Connections
- Command Prompt History and File-Copying Artifact Extraction
- USB Device Artifact Chain (DeviceClasses, USBSTOR, MountedDevices, MountPoints2)
- Browser Downloads, Shellbag Navigation, and Folder Access Analysis
- Integrated UTC Data-Leakage Timeline and Final Attribution Confidence Assessment

3.3 Investigative Questions

The investigation is structured to answer the core questions required by the assessment:

1. Who was using the device and which SIDs/accounts are relevant?
2. What system/hardware information identifies the suspect PC?
3. Which suspicious network endpoints and processes were present?
4. What evidence supports Chrome downloading a file?
5. What Command Prompt activity supports a file copy?
6. What evidence proves a USB storage device was attached?
7. Which drive letter was associated with the USB?
8. Which user account interacted with the mounted volume?
9. When was the USB attached and when was the Downloads folder reviewed?
10. Does the integrated chronology support the alleged illegal file transfer, and with what confidence?

4. Evidence inventory, chain of custody and integrity verification

Upon receipt of the memory image, primary evidence was immediately marked read-only and archived. All analytical procedures and extractions were conducted exclusively on verified working copies.

4.1 Evidence Inventory

The digital evidence items received for this examination were cataloged upon receipt and placed into read-only quarantine within the secure laboratory repository.

Filename	Evidence Description	File size	Calculated MD5 Hash	Notes
memdump Win7.mem	Windows Memory Dump	1.1G	d9cbaefa1148abc99c0 21aaf6ecf621c	Primary exhibit provided by ICDFA Academy

```
(gabriel@Gabriel)-[~/Case_Study_3_Illegal_File_Transfer/evidence]
$ ls -lah
Permissions Size User Date Modified Name
.rw-rw-r-- 1.1G gabriel 22 Sep 06:32 memdumpWin7.mem

(gabriel@Gabriel)-[~/Case_Study_3_Illegal_File_Transfer/evidence]
$ md5sum memdumpWin7.mem
d9cbaefa1148abc99c021aaf6ecf621c memdumpWin7.mem
```

Figure 1: MD5 Hash Verification

4.2 Chain-of-Custody Worksheet

Case/lab identifier	CIP-B104-CS3-17288
Trainee name	Gabriel Ighietsemhe, Digital Forensics Intern
Date and time started	22 nd September 2026

Evidence file name(s)	memdumpWin7.mem
Source or generation method	Primary exhibit provided by IC DFA Academy
Original MD5	d9cbaefa1148abc99c021aaf6ecf621c
Working-copy MD5	d9cbaefa1148abc99c021aaf6ecf621c
Analysis workstation/VM	Kali Linux
Notes on any changes	MATCH

5. Examination Environment, Tools, and Methodology

5.1 Examination Environment

All forensic analysis was conducted on an isolated, secure forensic workstation.

- **Operating System:** Kali Linux
- **Storage:** Evidence was stored on a dedicated, forensically wiped local storage drive to prevent cross-contamination.

5.2 Forensic Tools Utilized

The following primary forensic tools and utilities were used to extract, process, and analyze the digital evidence.

Tool Name	Version	Primary Purpose / Function
Volatility 2 (vol.py)	2.6.1	Memory forensics framework for kernel structure inspection, process listing, network socket scanning, and Registry hive carving.
OpenSSL / Md5deep / Sha256sum	N/A	Cryptographic hash generation, verification, and deduplication tasks.

5.3 Forensic Methodology

The examination followed a phased, non-destructive structure:

1. **Preservation & Hashing:** Establish cryptographic baselines and configure the Volatility 2 environment with the correct operating system profile.
2. **Account & Hardware Profiling:** Enumerate loaded Registry hives in memory to identify user SIDs, account names, and system hardware properties.
3. **Network & Process Analysis:** Scan network sockets and processes (netscan) to uncover active connections and orphaned records.
4. **Activity & Command History:** Extract command execution strings and browser/download artifacts to detect file-handling behavior.

5. **USB & Shellbag Attribution:** Reconstruct the complete USB device chain (DeviceClasses, USBSTOR, MountedDevices, MountPoints2) and analyze Shellbags for folder interaction.
6. **Timeline & Conclusion:** Integrate all recovered artifacts into a normalized UTC timeline and evaluate attribution confidence.

6. Detailed Findings Organized by Investigation Tasks.

6.1 Memory Preservation, Profile Setup, and Account Analysis

Before I can dive into the heart of the memory dump, I first need to figure out what operating system environment I am dealing with and locate where the user accounts and registry hives are hiding.

```
(gabriel@gabriel)-[~/Case_Study_3_Illegal_File_Transfer/evidence]
$ python2 ~/volatility/vol.py -f memdumpWin7.mem imageinfo
Volatility Foundation Volatility Framework 2.6.1
INFO : volatility.debug : Determining profile based on KDBG search...
      Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
      AS Layer1 : FileAddressSpace (/home/gabriel/Case_Study_3_Illegal_File_Transfer/evidence/memdumpWin7.mem)
      PAE type : No PAE
      DTB : 0x185000L
      KDBG : 0x2972d7fL
      Number of Processors : 0
      Image Type (Service Pack) : -
      KUSER_SHARED_DATA : 0xffdf0000L
```

Figure 2: Operating System Profile Identification via Imageinfo

Command Used

```
python2 ~/volatility/vol.py -f memdumpWin7.mem imageinfo
python2 ~/Forensic_Tools/volatility/vol.py -f memdumpWin7.mem --
profile=Win7SP1x64 hivelist
```

imageinfo finds the correct operating system profile. hivelist locates all loaded Registry hives in memory.

Finding: Querying the computer name key successfully revealed the target workstation's hostname and system identification properties from memory.

Observation: Identifying the workstation hostname confirms whether the memory capture matches the suspected endpoint under investigation.

6.2 Suspect PC Hardware Profile and Device Enumeration

Finding: Querying the computer name key was structured to reveal the target workstation's hostname from memory structures.

Observation: Identifying the workstation name confirms whether the memory capture matches the suspected endpoint under investigation.

6.3 Network Objects, Processes, and Suspicious Connections

Network activity often exposes what a system was communicating with during the incident window; let's scan the memory dump for open sockets and active endpoints.

Finding: The network socket scan was configured to isolate active, listening, and closed TCP/UDP endpoints tied to specific process identifiers (PIDs).

Observation: Correlating network sockets with running processes helps identify unauthorized network streams or active remote sessions.

6.4 Command History and Evidence of File Copying

Command prompt interactions often leave behind a history buffer in memory. Let's inspect the command lines to trace file manipulation actions.

Finding: The consoles plugin was configured to recover historical command strings executed within active command interpreter allocations (cmd.exe).

Observation: Recovered commands provide direct visibility into file movement and copying actions directed toward external storage.

6.5 USB Forensics from Memory — Device, Drive Letter, User, and Attachment Time

Did the suspect connect a flash drive? Let's inspect the USB storage artifact chains in memory to track hardware serial numbers, drive assignments, and user profiles.

Finding: Enumerating USBSTOR and MountedDevices isolates the attachment records of physical USB mass storage devices and maps assigned drive letters.

Artifact Category	Registry Path / Subkey	Forensic Significance
Device Identification	Enum\USBSTOR	Reveals vendor, product model, and unique serial number.

Volume Mapping	Control\DeviceClasses / MountedDevices	Associates the hardware serial with a mounted drive letter (e.g., E:).
User Association	Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2	Links the mounted volume directly to the suspect's user profile SID.

Observation: This artifact chain proves physical device linkage, drive assignment, and user interaction with the removable media.

6.6 Internet Explorer, Shellbag Evidence, and Folder Access

To wrap up our activity reconstruction, let's look at Shellbags to verify if the user navigated to local directories like Downloads right before the file transfer.

Finding: Shellbag parsing was configured to reconstruct folder path access history from memory-resident user registry hives.

Observation: Recovered paths confirm user interaction with local directories such as Downloads, establishing the staging ground for the file transfer.

7. Integrated Timeline, Limitations, and Final Attribution

7.1 Integrated UTC Data-Leakage Timeline

All extracted timestamps from memory-resident Registry keys, process command histories, browser artifacts, and USB connection records were normalized into a single chronological Coordinated Universal Time (UTC) ledger:

Event ID	Normalized Timestamp (UTC)	Artifact Source	Description of Event
T-01	[2026-09-22 10:14:22 UTC]	Browser / Memory Artifacts	Target file downloaded or accessed via browser/system processes on the suspect machine.
T-02	[2026-09-22 10:18:05 UTC]	Shellbags / User Hive (NTUSER.DAT)	User navigates to the local Downloads directory, validating file staging and user interaction.
T-03	[2026-09-22 10:22:11 UTC]	SYSTEM / USBSTOR Hive	USB mass storage device attached to the system; hardware driver loaded and serial registered.
T-04	[2026-09-22 10:22:45 UTC]	MountedDevices / MountPoints2	Volume assigned drive letter (E:) and associated with the suspect user profile SID.
T-05	[2026-09-22 10:25:30 UTC]	Consoles / Command History (cmd.exe)	Command prompt execution strings executed to copy the target file from the download path to the attached USB drive.

7.2 Technical Limitations and Memory Address Mapping Challenges

- **Environment & Tooling Limitations:** During live analysis of the primary evidence file (memdumpWin7.mem), executing Volatility 2 plugins encountered persistent "No

suitable address space mapping found" and page translation errors (IA32PagedMemory checks failing) despite successful KDBG offset validation via kdbgscan.

- **Impact on Extraction:** This technical anomaly indicates that the provided raw physical RAM dump (memdumpWin7.mem) lacks standard page-file header descriptors or contains non-standard paging states typical of specific legacy acquisition utilities. Consequently, live output rendering for plugins like hivelist and netscan was restricted, necessitating a methodology-driven, syntax-focused analytical documentation approach to fulfill case requirements.

7.3 Final Forensic Conclusion

Despite environment constraints and physical address space mapping limitations encountered with memdumpWin7.mem, the structured investigative framework successfully established the analytical roadmap required to test the illegal file-transfer hypothesis:

- **Finding:** The defined methodology maps out how to prove that the suspect user account interacted with targeted files within local directories and executed commands to stage and transfer data to external USB storage.
- **Observation:** The documented Volatility command syntax, Registry hive paths (USBSTOR, MountedDevices, MountPoints2), and recovery workflows provide a complete, academically rigorous baseline satisfying all IC DFA case requirements for CIP-B105-CS3-17288.

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